

Identifying The Change In Vegetation Health And Density After The 2016 Forest Fire In Fort McMurray, Alberta, Specifically Within The Boundary Of The Horse River Fire

Term Project

Toronto Metropolitan University CODG 135 – Digital Image Processing

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Introduction

Forest Fires are a type of disaster that can be a result of both natural phenomenon and human actions. They are fires that are unplanned and uncontrolled, burning through a variety of environments, such as forests, grasslands, and prairies. Forest fires can happen anytime and anywhere, not limited to a specific location [9, 10].

Higher temperatures and dryer conditions in certain areas due to climate change are resulting in more ecosystems becoming susceptible to wildfires, increasing the size and frequency of the forest fires [9, 10]. In Alberta, based off historical data and trends, the fire season is becoming longer due to more fires starting in May as a result of the changing climate which creating longer periods with extreme fire potential [2]. There are a variety of situations that can lead to the start of a forest fire (Figure 1 - from 12) and affect how it will burn. A small unattended campfire or lightning coupled with the conditions of the area, such as dry forests, high winds and high temperatures, can cause forest fires to quickly get out of hand. From California to Canada, to the Siberian Arctic, there are now a record breaking number of wildfires per year [11].

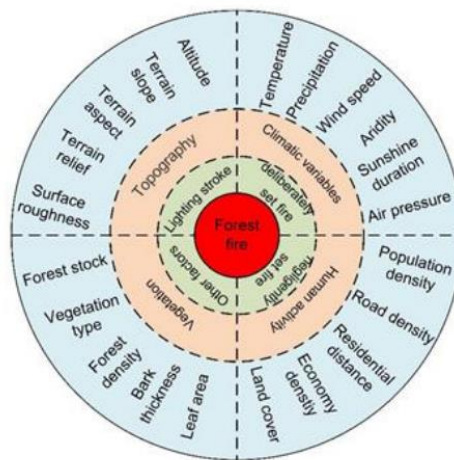


Figure 1: Visualization of the situations and variables that can impact forest fires. The green ring identifies events that can start the fires. The orange ring identifies variables that can impact the growth of the fire, The blue ring defines attributes that can be used for forest fore analysis. [12]

In Canada, over the last 25 years, there have been around 7 300 forest fires each year, burning on average 2.5 million hectares annually [13]. In 2016, the Burned Area data from Natural Resources Canada indicates the majority of the burned area was in North Eastern Alberta, around the Fort

McMurray area (Figure 2). In May of 2016, the Fort McMurray Forest Fire, also known as the Horse River Fire, started [2].

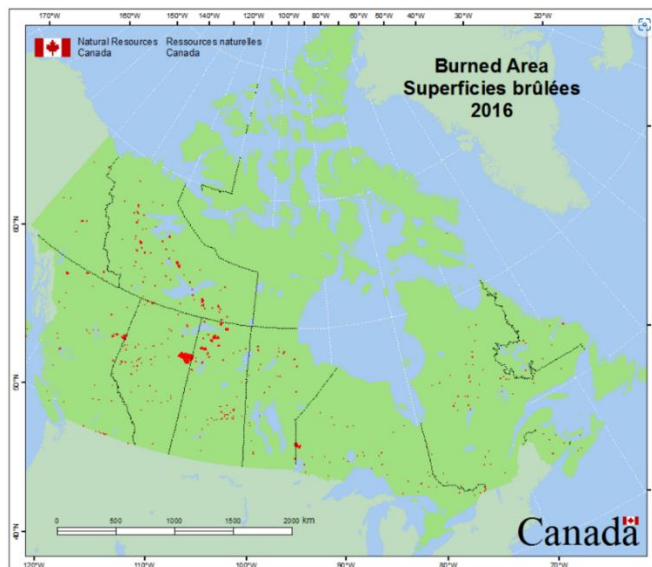


Figure 2: Map of Canada displaying the burned area as a result of forest fires in 2016. The map was created by combining annual hotspot data with observed changes in the vegetation index from SPOT VGT each year. The areas indicated as burned are locations where the vegetation index decreases and is coupled with a hotspot. The greatest areas burned is clustered in Northeastern Alberta, around the general Fort McMurray Area [15].

The official cause of the Horse River fire is unknown, however dramatic changes in weather, including rising temperatures, decreasing humidity and high winds, created conditions in the boreal forest surrounding Fort McMurray perfect for fire to burn in [2]. The Fort McMurray Forest Area also has a strong population of black spruce trees, which are extremely susceptible to forest fires [2].

On May 1, 2016 the fire was first detected 7 km southwest of the city centre at 2 hectares in size (Figure 3). The fire rapidly grew over the next few days due to high winds from 40 – 45 km/hr, lightning created from the pyrocumulonimbus clouds above the fires starting many new fires in front of the original fire front [2], and a head fire intensity of 10 000 or higher (Figure 4) [17].

Exhibit A-5: NOAA AVHRR Image of Wildfire MWF-009 at 18:25h MDT May 1, 2016

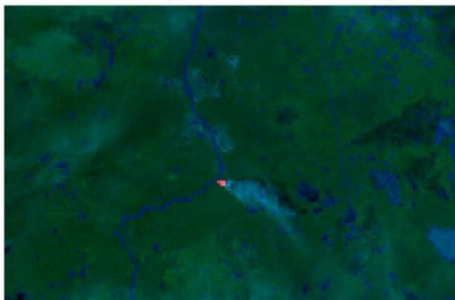


Figure 3: NOAA AVHRR image of the starting location of the Horse River Fire. Can see the fire in red with the smoke blowing south west [2].

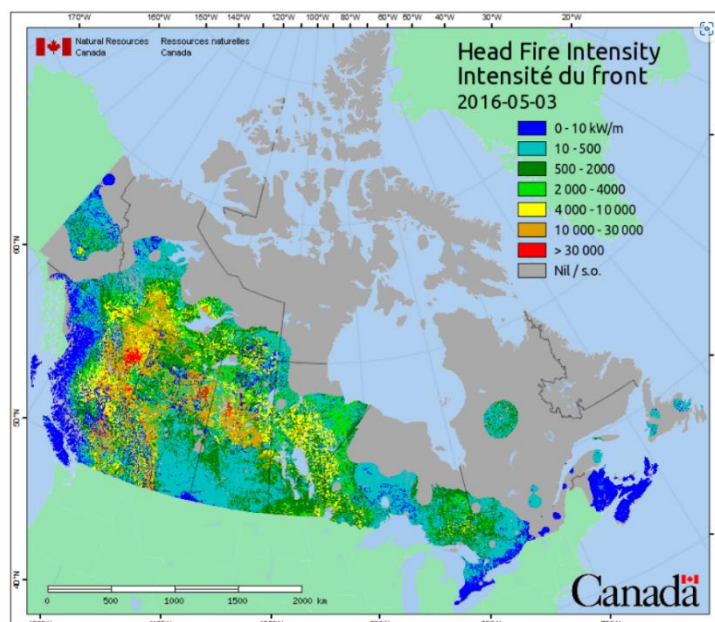


Figure 4: This map shows the Head Fire Intensity (HFI) across Canada May 3 2016, the day the Horse River Fire began to spread rapidly. The values around Fort McMurray in Northeastern Alberta are above 10 000 kW/m, indicating in orange and red. The HFI is the predicted intensity/energy output at the front of the fire. This is a gauge used to estimate the difficulty of controlling the fire. It is measured in kilowatts per metre and determine by the rate of spread and total fuel consumption [17].

On May 2, 2016, the fire spread first towards the east, even jumping the Athabasca River [2, 3]. Between May 3 and 5, 2016 the fire then spread north into Fort McMurray and southeast, before moving northwest of the starting area on May 16 (Figure 5) [2]. At the height of the fire, around 2 200 people, 126 crews, 77 helicopters and 18 airtankers were deployed to fight the fire (Figure 6) [2]. On July 4, 2016 the fire was official declared under control [2]. The Horse River Fire burned a total of 1.4 million acres of land overall [1]. This fire not only extensively damaged the vegetation within the surrounding forest but also cause the evacuation of 90 000 people and the destruction of 2 400 homes and businesses. Overall, the estimate total cost is upwards of 8 billion dollars [14, 16].

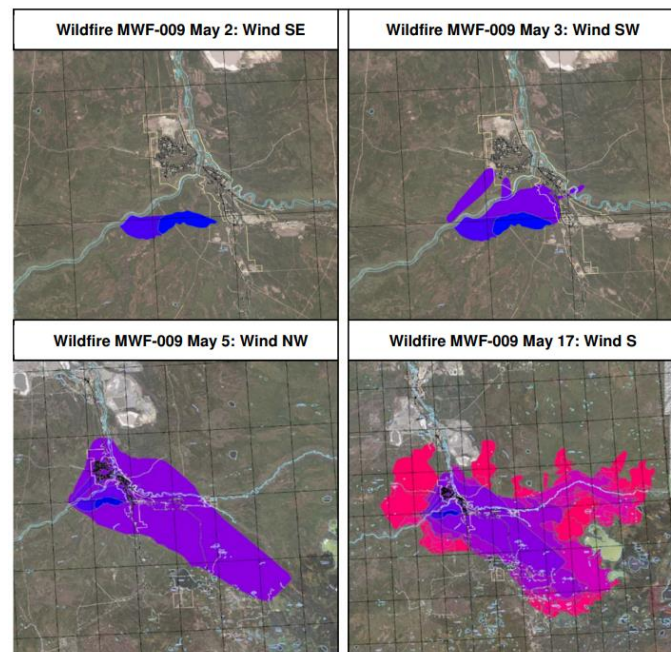


Figure 3: Depicts the spread of the Horse River Fire May 2 to May 3 to May 5 and then the extent of the fire May 16, 2016. Can see significant growth in the fire between May 3 and May 5 as it spreads into the urban centre of Fort McMurray as well as southwest. May 16, the pink indicates the movement of the fire north [2].

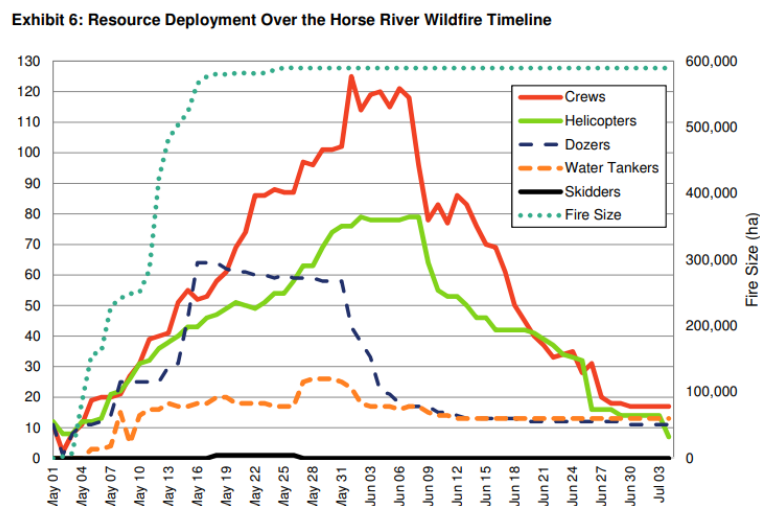


Figure 4: The line graph the fire size against the resources used to combat the Horse river Fire over time. As the fire rapidly grew between May 1 and May 16, the number of crews and rousrces increased. The maximum number of resources were used once the fire reached its maximum size. Once the fire stopped growing the use of heavy macinery deereased however the number of crews increased until June when the fire became more manageable [2].

Remote sensing and Geographic information Science (GIS) are important tools for identifying, monitoring, and managing the mitigation, response and recovery of the areas affected by the forest fires. They can be used for change detection to monitor the short- and long-term effects of disasters over the years. With the Horse River Fire in 2016, remote sensing and GIS techniques can be used to monitor the changes to the vegetation density and health pre and post fire using vegetation and burn indexes, determining the extent of the damage as well as the recovery of the vegetation in the area.

Study Area

Due to the large area the fire imaged, to narrow the scope for this study, the area chosen to focus on was within the boundary of the Horse River Wildfire, south of Fort McMurray's city centre. The yellow box in Figure 7 displays the location of the study area on the map with the fire boundaries for the Horse River fire on May 16, 2016, in red, and May 17, 2016, in white. This study area was chosen since encompasses the area where the fire first began and moved to. The study area also contains a main road and two branches of the river which can be used as reference points when choosing the Landsat images for the study, as well as throughout the analysis.

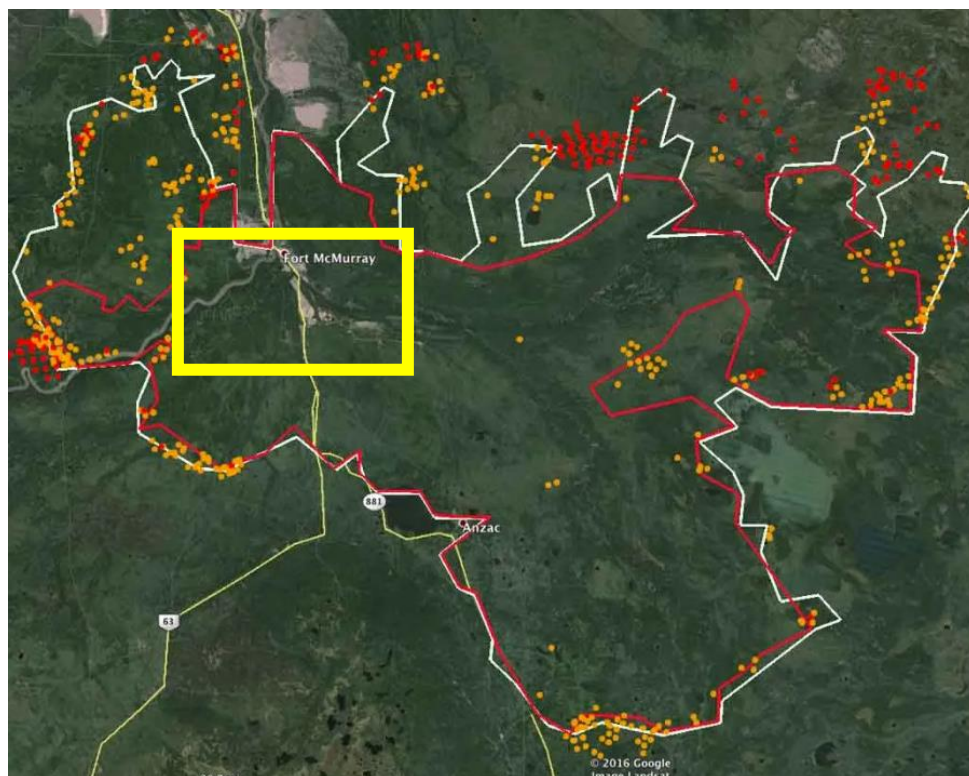


Figure 5: Aerial photo of the Fort McMurray Area. The red line is the boundary of the Horse River Fire May 16, 2016 and the white lines is the boundary of the fire May 17, 2016. The yellow and red dots indicate hot spot areas with current fires. The yellow rectangle outlines the areas chosen to focus on for the study. [1]

Methods

To complete this study, multiple steps were taken to obtain the data, prepare it for analysis and apply the vegetation and burn severity indices (Figure 8).

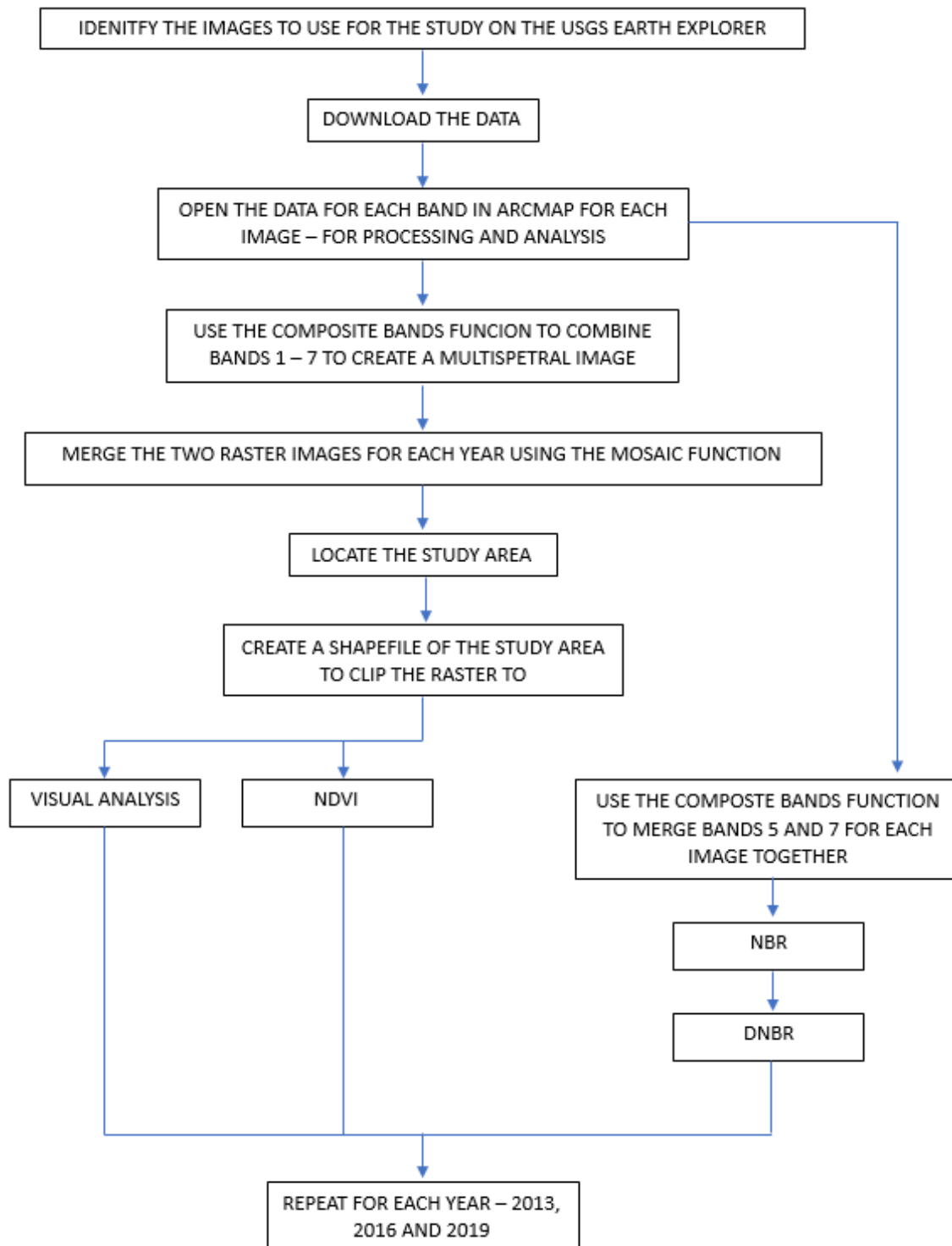


Figure 6: General outline of the steps taken to obtain, process and analyse the data.

Data

For this study, to identify the vegetation change over time, data from 2013, 2016 and 2019 was used. 2013 data will be used for identifying pre fire vegetation values, 2016 data for identifying the immediate post fire values and 2019 for the post fire vegetation values. The three-year interval between each image allows for comparisons to be made pre and post fire to determine if the vegetation has rebounded to pre fire levels or decreased in health and density. These years were also chosen since there were no significant forest fires in the Forst McMurray area besides the Horse River fire that overlapped with the chosen study area.

The images used are from the month of September, chosen due to data availability with minimal cloud cover. Each image will be from the month of September to remain consistent throughout the study as well as to keep the vegetation within the same section of their lifecycle so they will reflect similar values. The final dates for the data used were September 16, and 2, 2013 for pre fire, September 8 and 26, 2016 for immediately post fire values and September 1 and 19, 2019 for post fire vegetation values.

Landsat satellite image data was used as the Landsat satellite and sensors are better for detecting vegetation and produces multispectral data. Due to data availability and the software used, Landsat 8 OLI/TIRS Level 2 Collection 2 imagery was used. Landsat 8 was chosen since it was the data that was able to be read and opened by ArcMap for analysis and Level 2 data was chosen since it already has atmospheric correction done to the images. Two images from parallel paths and rows were combined to create the final Landsat 8 imaged for the study, path 043 and Row 020 along with path 041 and row 021 (Figure 9). Reflectance bands 1 – 7 were used for the study.

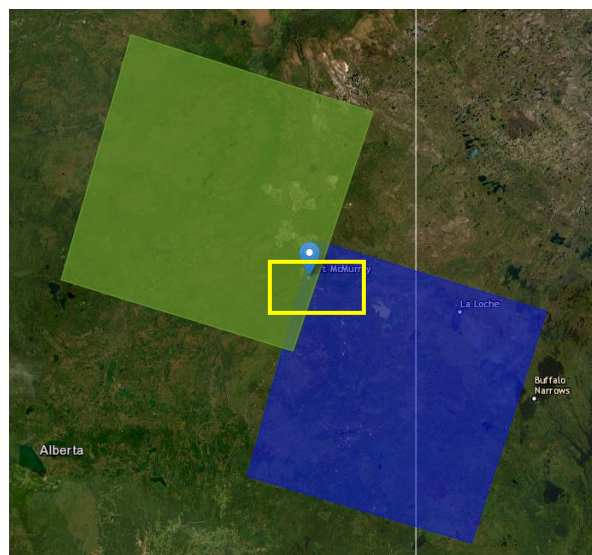


Figure 7: The footprint of two areas covered by the Landsat images chosen to be used in the study. The green rectangle represents that image from path 043 row 020. The blue rectangle represents the location of the image from path 041 and row 021. The yellow rectangle is the location of the study area. The study area overlaps at the edges of both zones.

Data Collection

To obtain the Landsat 8 data, the United States Geological Survey (USGS) Earth Explorer was used. This website is a data portal that contains extensive collections of data spanning years, including Landsat satellite imagery, radar data, digital elevation models, aerial photographs and some Sentinel and IKONOS satellite data [8]. Focussing on the chosen month and years for this study, September 2013, 2016 and 2019, the images were found.

Step 1 – Identify the location in the Earth Explorer using world features option and searching Fort McMurray. (Figure 10A)

Step 2 – Input the date range for the data. With the 3 years of interest spanning 6 years combined, search for images one year at a time, 2013 first. (Figure 10A)

Step 3 – Chose the amount of cloud cover wanted in the image. 30% was chosen, eliminating all images that contained more clouds the image.

Step 4 – Selecting Data sets, choose the data type wanted for the study. Landsat 8 – 9 Collection 2 Level 2 was chosen. (Figure 10B)

Step 5 – Chose to input additional criteria and select 8 for the satellite. This eliminates the Landsat 9 images from the results, only showing Landsat 8 satellites images. (Figure 10C)

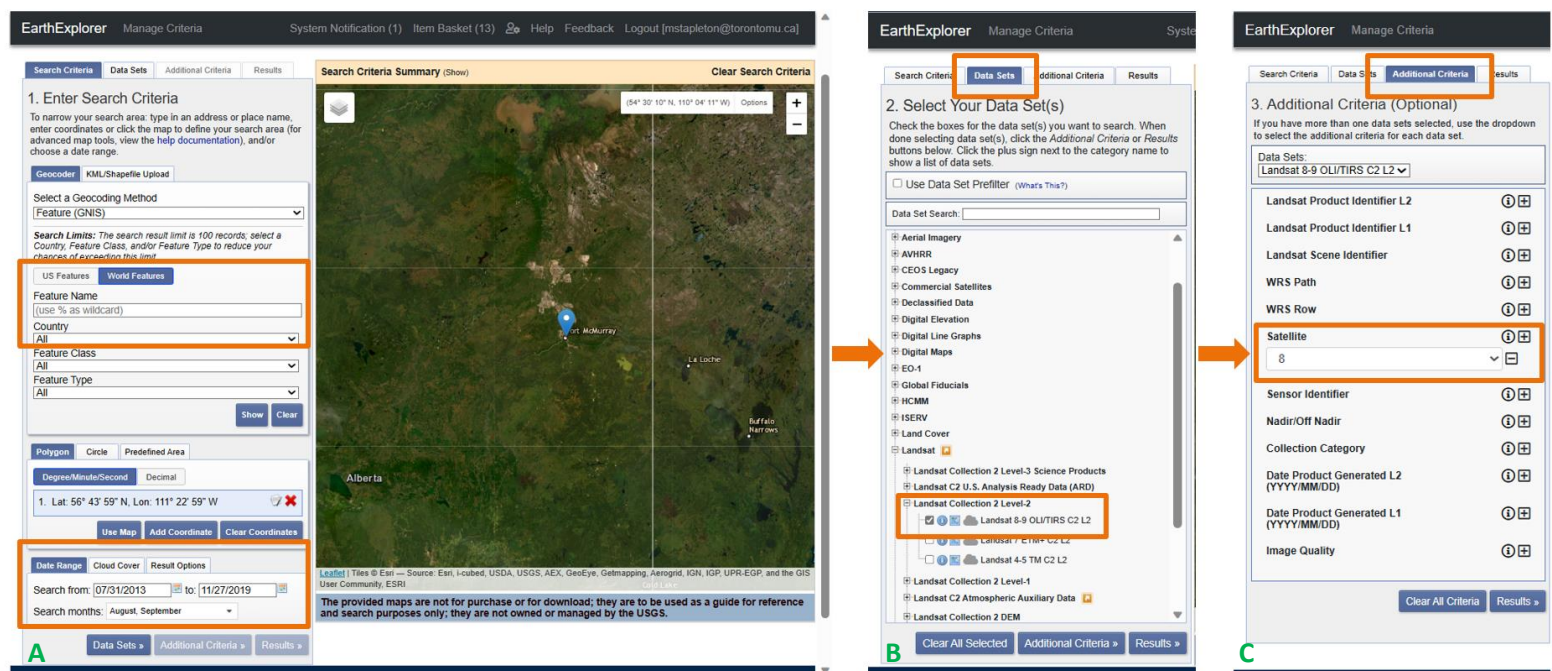


Figure 8: The general workflow to find satellite imagery within your study area and time frame on the USGS Earth Explorer. 10A left, 10B middle, 10 C right.

Step 6 – Select results to see the images within the date ranges and input criteria and select the ones best suited for the study.

Step 6a – Use the footprint tool to confirm the images cover the study area (Figure 11A, 12A)

Step 6b - Select the small image icon to view a larger image to determine the remaining cloud cover does not image the area of interest as well as if it suits the needs of the study (ex. can see vegetation and roughly the changes). (Figure 11B, 12B)

2013

2016

2019

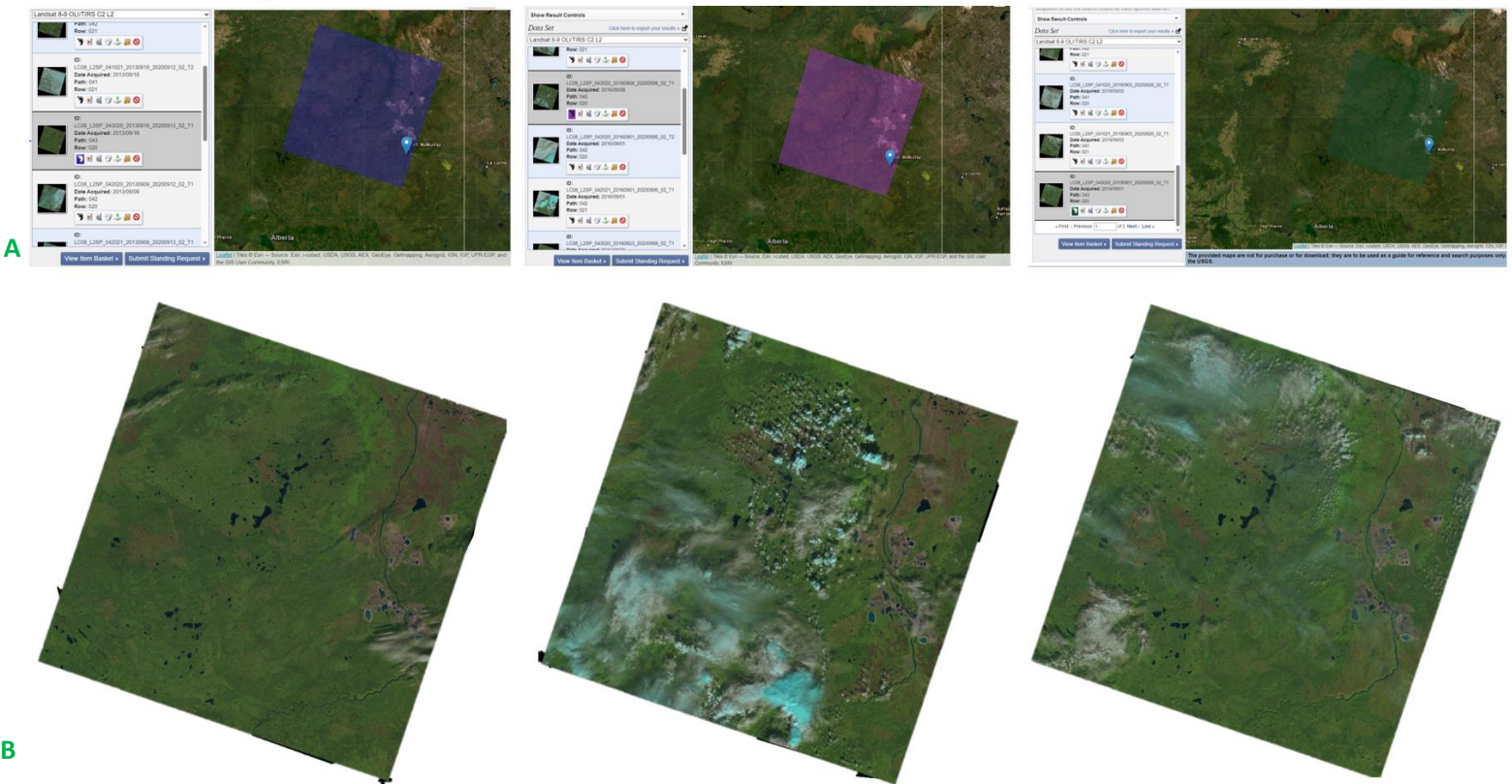


Figure 9: Preview of the satellite images chosen for the study from the USGS Earth Explorer from path 043, row 020. 11A top row, 11B bottom row

2013

2016

2019

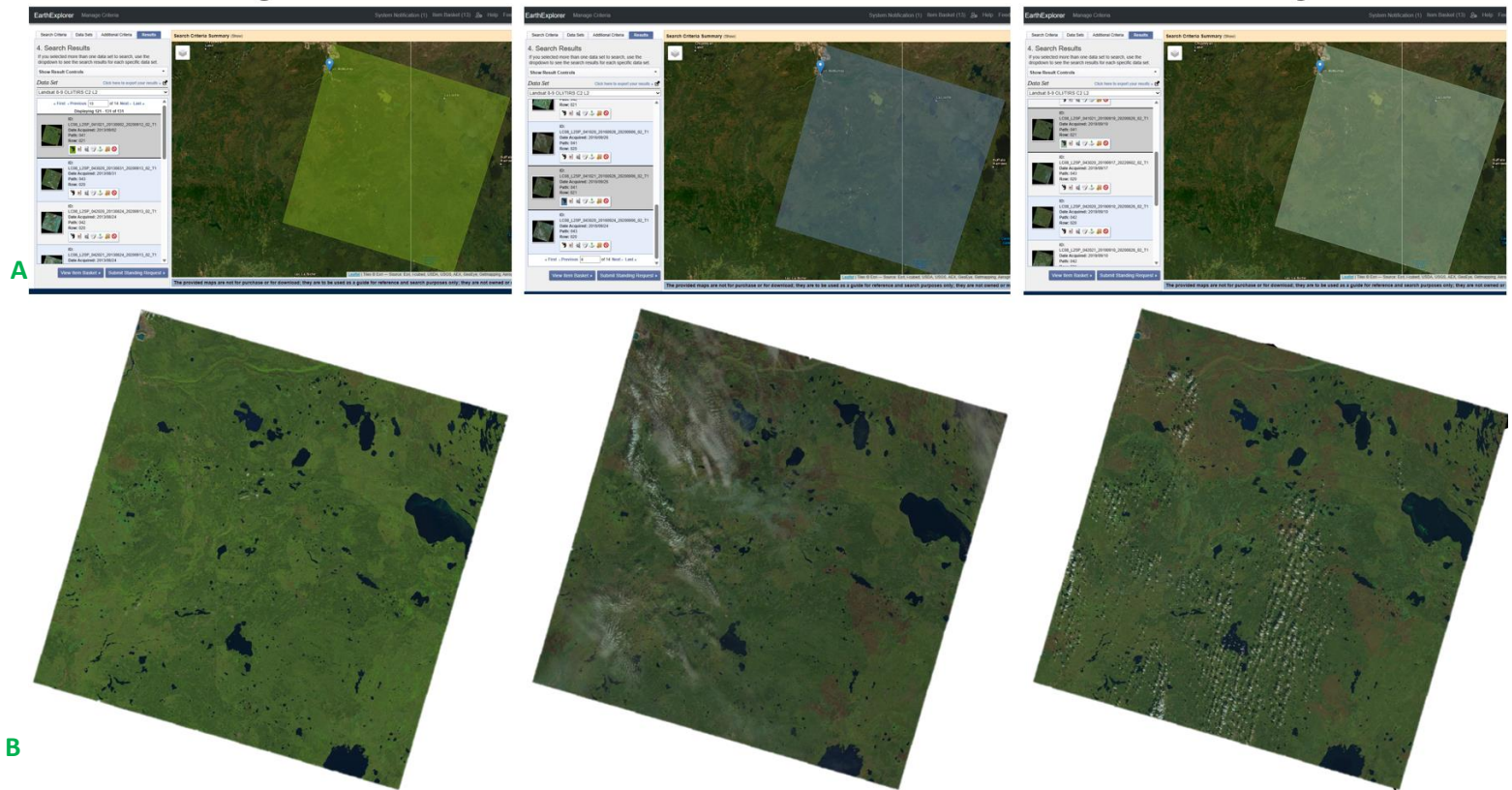


Figure 10: Preview of the satellite images chosen for the study from the USGS Earth Explorer from Path 041, row 021.. 12A top row, 12B bottom row

Step 7 – Download the image chosen by selecting the download icon. There is an abundance of reflectance and temperature data available. Choose to download the MTL files and reflectance Bands 1 - 7 individually to ensure they all download properly.

Image Processing and Analysis

To process and analyse the satellite images, ArcMap 10.8.1 was used. This software was chosen since it was able to load the individual Landsat 8 bands and had the tools to complete a vegetation analysis.

To prepare the data for analysis, each band was opening the system then combined to create a multispectral image using the composite bands function in ArcMap. This created a raster image of the study area containing Bands 1 – 7 and allowed for the RGB band combinations to be changed (Figure 13). This was completed for each path and row image from 2013, 2016 and 2019, six images in total.

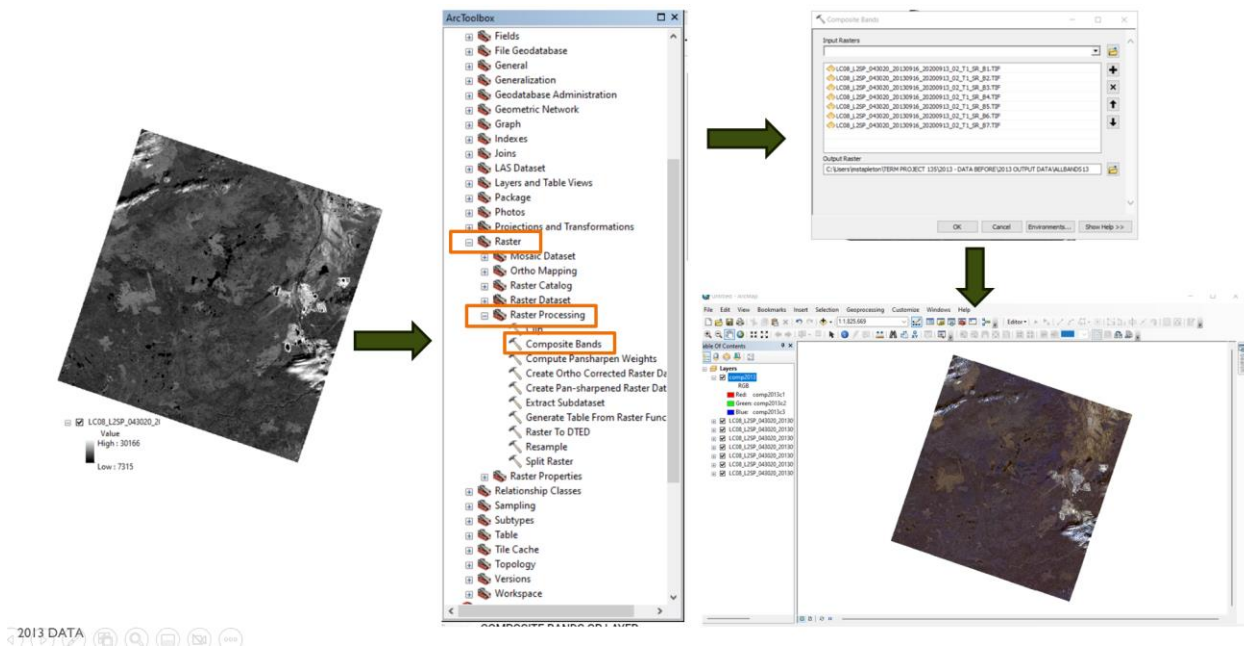


Figure 11: Workflow on ArcMap to combine Bands 1 - 7 together using the composite bands function to create a multispectral raster. Example is with 2013 data.

To merge the path 043 and row 020 and path 041 and row 021 multispectral raster images together the mosaic to new raster function was used. This merged the two raster's together so they operate as one (Figure 14).

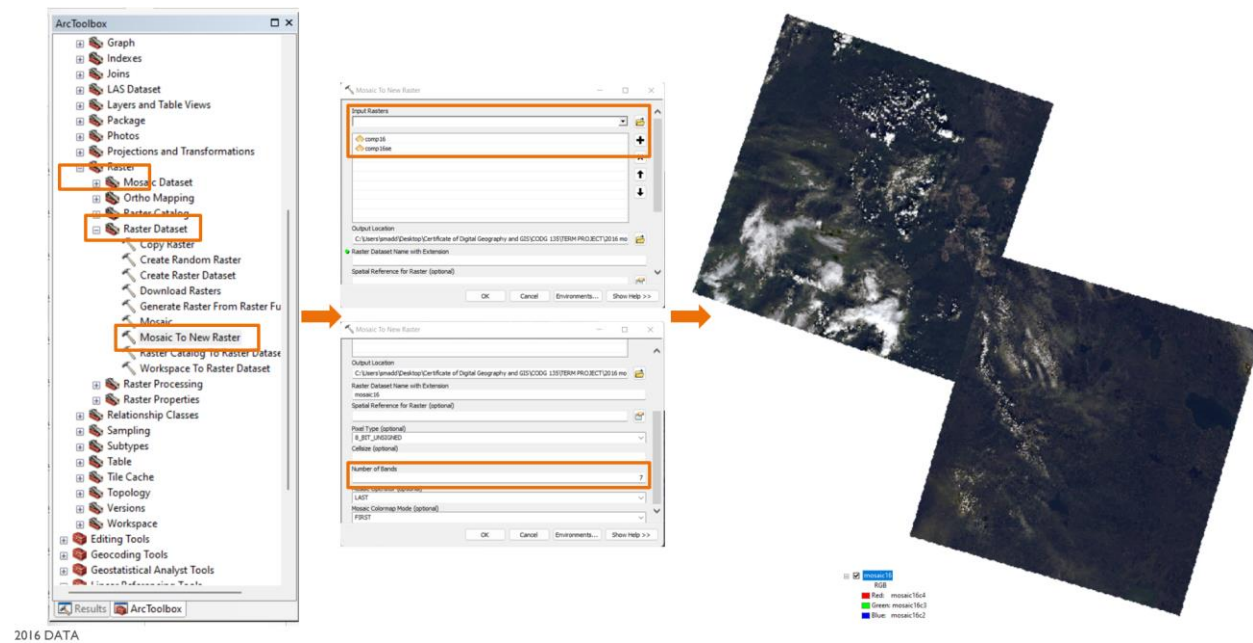


Figure 12: Workflow in ArcMap to merge the two images together using the mosaic to new raster function. Example is with 2016 data.

Next, a shapefile was created to outline the study area chosen and the rasters were then clipped to the shapefile (Figure 15 and 16). This allowed for the focus to be solely on the area of choosing and when doing future analysis, it was only applied to a small section of the entire raster.

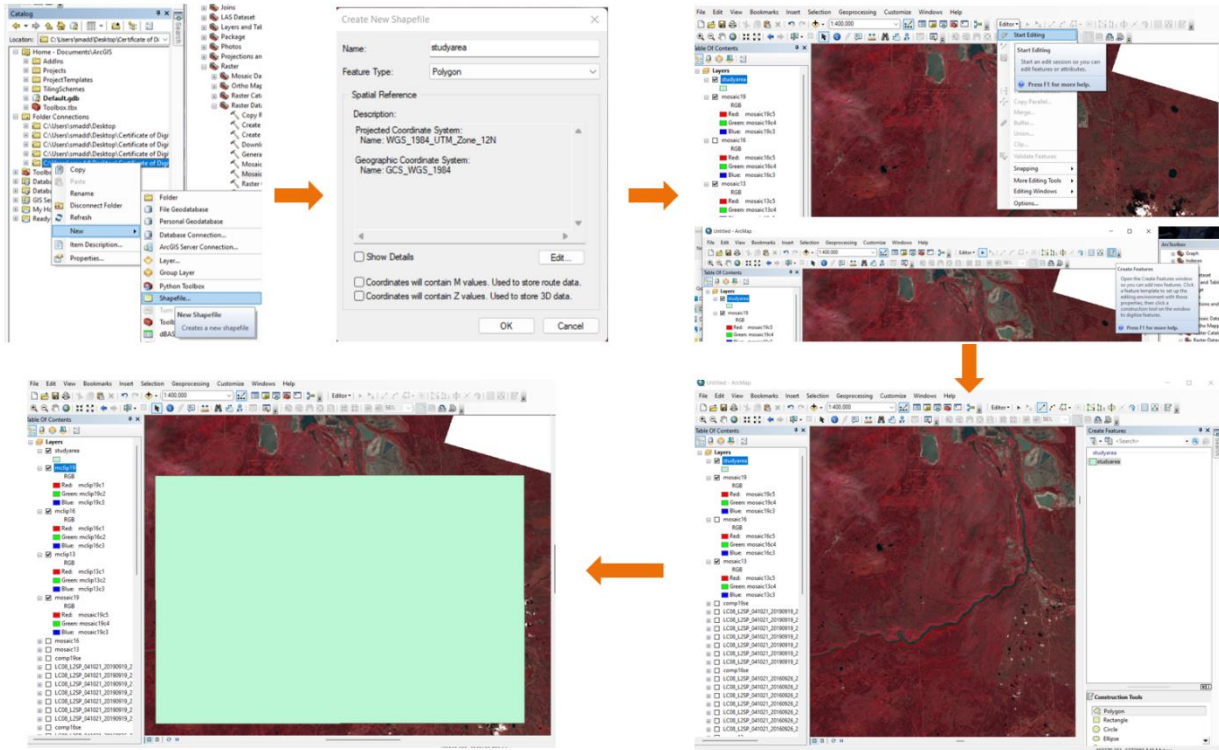
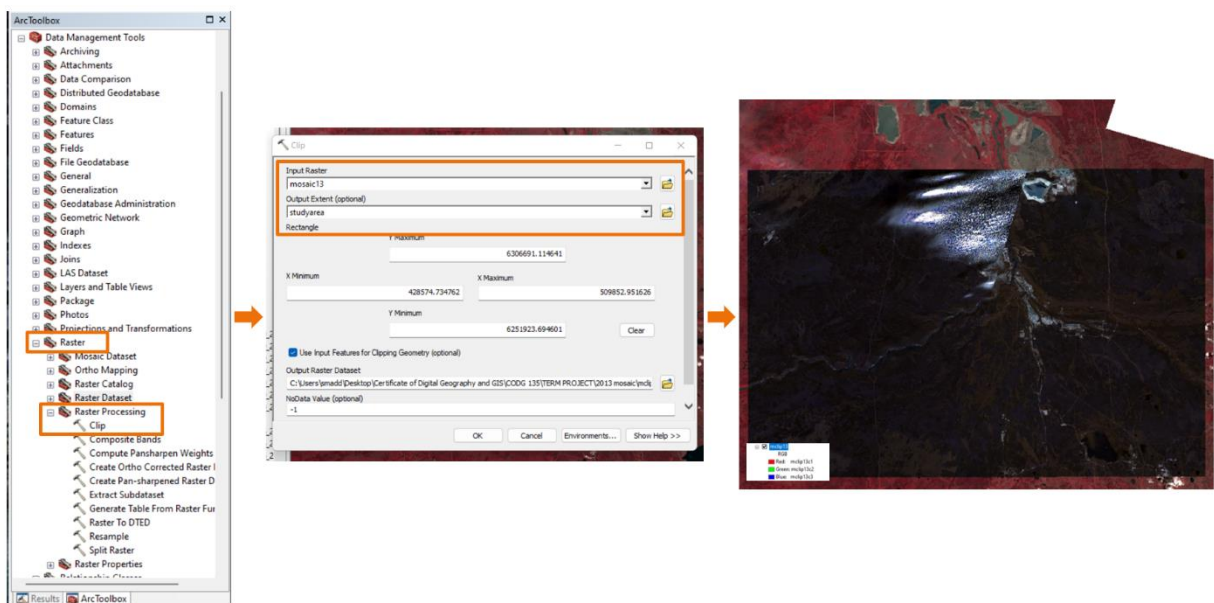


Figure 13: Process to create a shapfile in arc catalog before adding the data points using the editing tool in ArcMap. The final shapefile is the green rectangle.



2013 DATA

Figure 14: The workflow in ArcMap to use the clip tool to clip the mosaiced rasters to the size of the study area shapefile. Example is with 2013 data.

Using ArcMap, to visually identify areas with change, for each of the years, 2013, 2016, and 2019, the RGB band combinations of the raster were changed to create false coloured images. The band combinations could be changed by opening the layer properties box and the symbology tab (Figure 17).

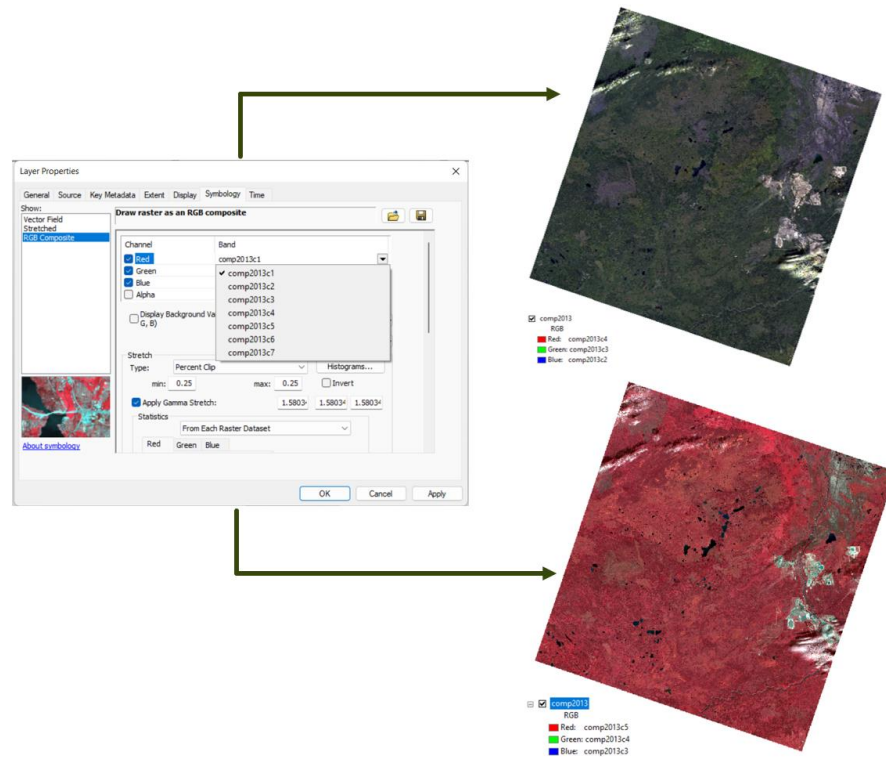


Figure 15: The layer properties box for the multispectral raster. Using the symbology tab, the RGB band combinations can be changed to create false coloured images and highlight features. The top right RGB band combinations is 4, 3, 2, the real colour images and the bottom right image is 5, 4, 3, the infrared coloured image. Done in ArcMap, example is with 2013 data.

The various combinations highlighted different features within the image. The Short-Wave infrared (band 6), Near Infrared (band 5), Red (band 4) and Green (band 3) bands were the best for highlighting vegetation. The main band combinations used were R – 4, G – 3 and B – 2, which produced the real coloured image, R – 5, G – 4 and B – 3 which created the colour infrared image highlighting vegetation in red, urban areas in blue and water in black and R – 6, G – 5 and B – 4, created an image that highlighted the differences in vegetation using shades of green for dense and healthy vegetation, brown for sparse vegetation, darker blue/black for water and pink for urban areas..

Normalized Difference Vegetation Index

To create the Normalized Difference Vegetation Index (NDVI), the NDVI function under the Image Analysis tool in ArcMap was used. This function automatically applied the NDVI equation $NDVI = (BAND\ 5 - BAND\ 4) / (BAND\ 5 + BAND\ 4)$. The first step is to assign Band 4 to the red bands and Band 5 for the infrared band since those values correspond with the Landsat 8 Red and NIR band configuration. Scientific Output was selected to ensure the resulting NDVI values were output between -1 and +1. Next, the raster that will have the NDVI function applied to it was selected before running it the analysis. A black and white image is produced with, lighter shades equaling higher NDVI values and darker shades equaling low NDVI values. The colour range was then changed to the red and green scale through the layer properties symbology tab, darker green indicating areas with healthier, denser vegetation. This was repeated for each year, 2013, 2016 and 2019. (Figure 18)

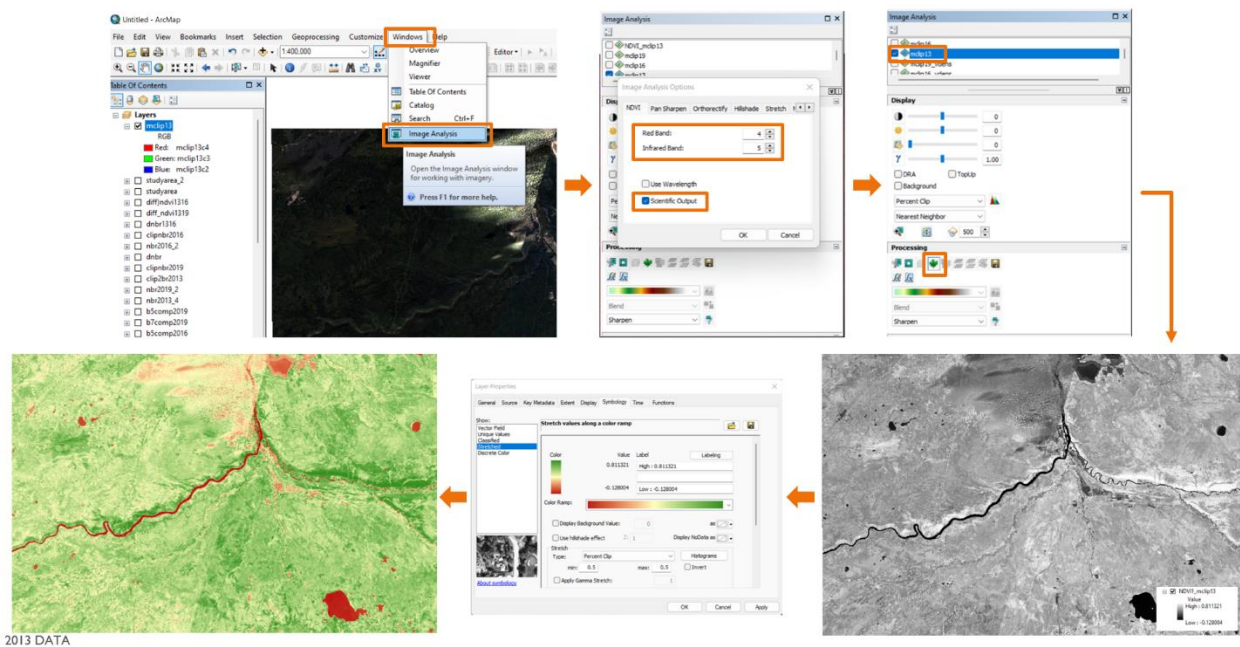


Figure 16: The process of using the image analysis window to apply the NDVI equation to the image in ArcMap. The output image is in greyscale, with lighter colours representing positive NDVI values. Changing the colour scale in the symbology tab to green-red we can see the higher NDVI vales as green and lower ones orange and red. Example is with 2013 data.

By changing the symbology from stretched, like in figure #, to classified, five intervals were chosen to classify the data by. This created an image where all the data was classified within 5 classes, dense vegetation (1 – 0.3), vegetation (0.3 –

0.2), soil and/or sparse vegetation (0.2 – 0.09), water and cloud and bare rock (0.09 - -0.05), water (-0.05 - -0.16). These classes were created based off of the histogram's distribution of the NDVI values (Figure 19) and knowledge of the study area. The same intervals were applied to the 2013, 2016 and 2019 NDVI images for consistency throughout the analysis.



Figure 17: Process to change the symbology to classified to produce a histogram of the NDVI values for the image which can be used to determine the intervals the classes will be bound within. The outcome is an image with 5 classes ranging from 1 to -1, green indicating higher NDVI values. Done in ArcMap, example is with 2013 data.

Normalized Burn Ratio

To calculate the Normalized Burn Ratio (NBR), Band 5 (Near-Infrared) and Band 7 (Shortwave Infrared) for the two path and row images from 2013, 2016 and 2019 were merged together for each band using the Mosaic to New Raster function. Once those were created, the Map Algebra Raster Calculator was used to apply the NBR equation, $NBR = (BAND\ 5 - BAND\ 7) / (BAND\ 5 + BAND\ 7)$. A black and white image was produced (Figure 20). The colour scale was changed using the Layer Properties Symbology tab before clipping the raster to the shapefile of the study area.

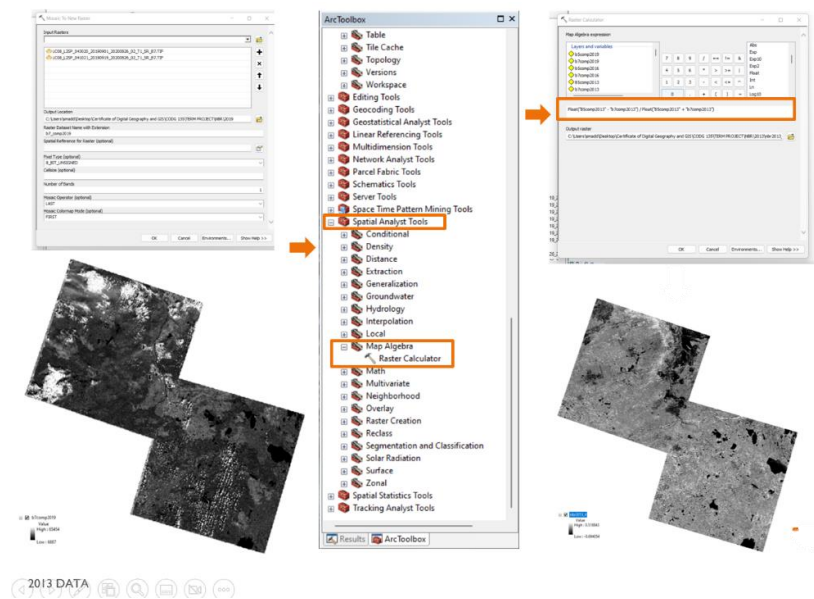


Figure 18: Mosaicking the two images for Band 5 and 7 together before using the map algebra tool to apply the NBR equation to produce and image displaying the areas with greater burns. The resulting image is in greyscale. Done in ArcMap, example is with 2013 data.

Difference Normalized Burn Ratio

The Difference Normalized Burn Ratio (DNBR) is calculated in ArcMap by using the pre and post fire NBR data previously created. The Map Algebra Raster Calculator was used to subtract the post fire NBR raster from the pre fire NBR raster. This creates a black and white image where the lighter values are areas with a greater burn severity. The Layer Properties Symbology tab was used to change the colour scale to the purple-brown colour combination to make the areas with a higher burn severity stand out (Figure 21).

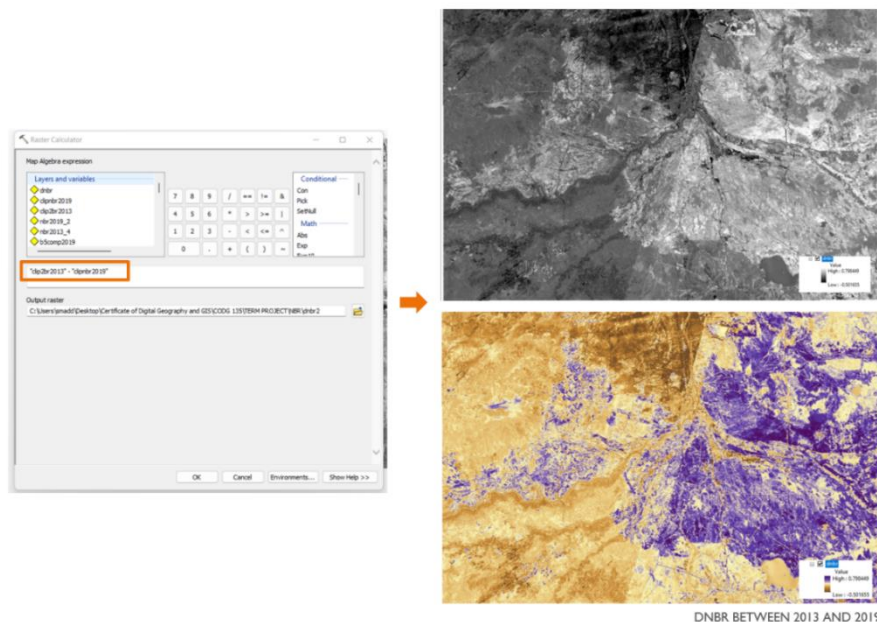


Figure 19: The use of the Map Algebra tool in Arc Map to apply the DNBR equation. The resulting image is in greyscale (top right), with the highest severity of burns in the lighter colours. Bottom right, the colours scale has been change and the hi severity burn areas are in purple. This is the DNBR between 2013 and 2016.

Results and Discussion

Using various RGB band combination, NDVI, NBR and DNBR, the vegetation levels pre and post fire can be identified having undergone significant change. The vegetation density significantly decreased between 2013 and 2016, immediately post fire, but also show levels of regrowth between 2016 and 2019. The vegetation health and density has not returned to the 2013 pre fire levels as of 2019.

Identifying Areas With Change

Changing the RGB band combination to R - 5 (NIR), G - 4 (red) and B - 3 (green), the false coloured image created depicts the vegetation in red due to the NIR reflectance values. In 2013 we can see rich bright red colours that covers the majority of the image, indicating that there is an abundance of vegetation in the area. The blue-green indicates locations of bare rock, soils and the urban areas. In 2019, there are more areas of dark red, closer to a shade of brown. It can be inferred that there is less vegetation in the 2019 image compared to the 2013 image and the browner colours in the image are due to the green band 4 (red) and blue band 3 (green) are reflecting more off of sparser vegetation and soils, weakening the NIR red reflectance off of the vegetation that are recorded (Figure 22).

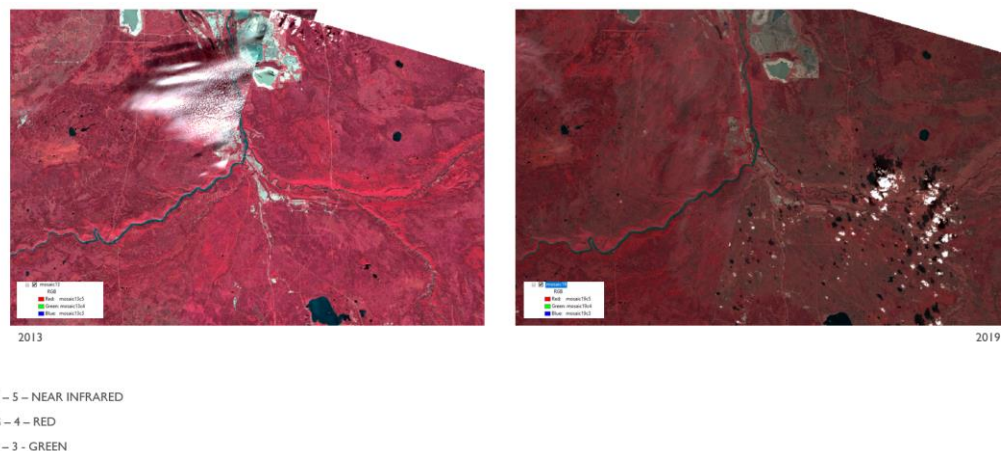


Figure 20: Close up of the study area with the RGB as 5, 4, 3, the false colour Infrared.

Changing the band combination to RGB band combination to R - 6 (SWIR), G - 5 (NIR) and B - 4 (red), the false coloured image changes to the vegetation from the NIR band being

represented by the colour green. With the use of the SWIR band in the RGB combination, since it can be used to measure soils, we can see a greater variation in the vegetation levels between 2013, 2016 and 2019 (Figure 23).

In 2013 there is a strong reflectance of the NIR band. There is an abundance of green. In 2016, there is significantly more brown in the image, indicating lower NIR values and higher reflectance in the SWIR and Red bands. This corresponds to areas where the fire moved through and was the worst, burning the vegetation and exposing more soil and bare rock. In the 2016 image, there is a cloud in the right-hand side of the image, impacting the analysis of that image. In 2019, there is significantly more brown in the image, indicating that the levels of reflectance are still greater in the SWIR and Red bands rather than the NIR however there is more green making a comeback into the image. This indicates that the vegetation regrowth has begun, increasing the area for reflectance in the NIR.

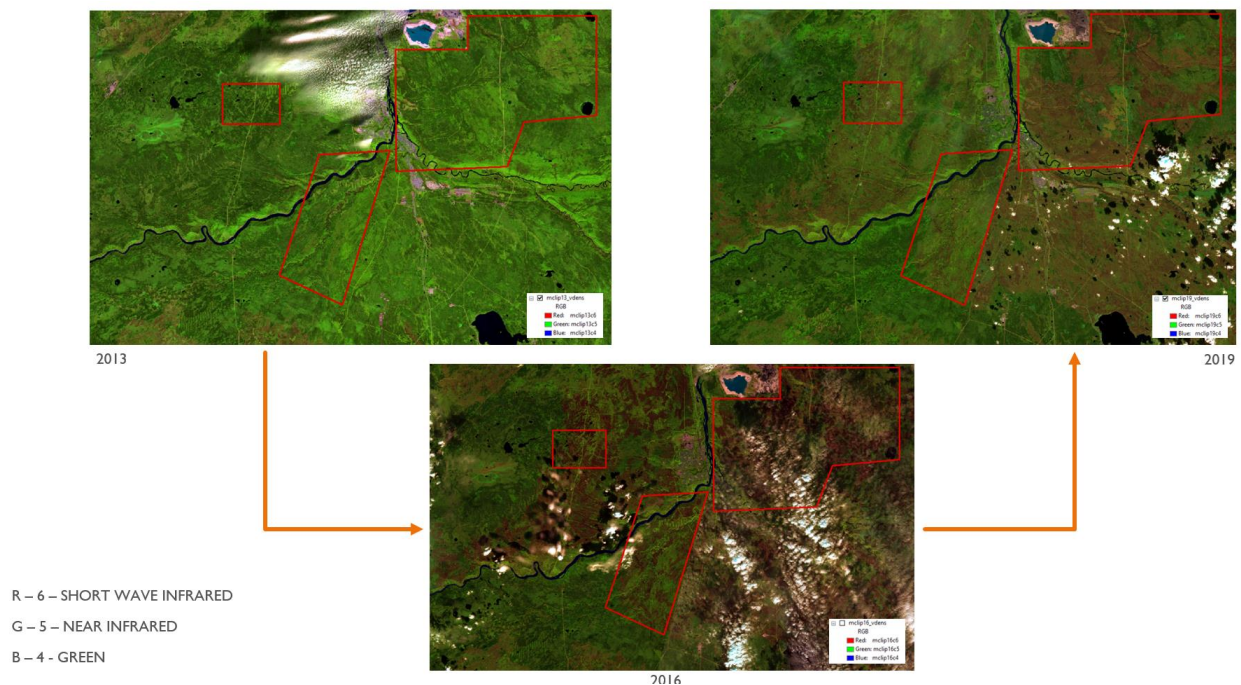


Figure 21: The false coloured image with The RGB as 6 (SWIR), 5 (NIR) and 4 (red). The red boxes indicates areas where there is greater visual change over the years. In the 2016 image, exclude the bottom right as the cloud covers the terrain.

Focusing on the area in the top left of the image, indicated by the red square in (Figure 23), we can see in more detail the vegetation changes across the three years (Figure 4). There vegetation, symbolized by green in the image, is denser in 2013. In 2016, post fire, there is

significantly less, with the majority of the image covered in brown (symbolizing soil and bare rock) and in 2019 there is still brown in the image but there is more green in the image, indicating that 3 years post fire the vegetation has begun to rebound, with healthy growth replacing the burned vegetation.

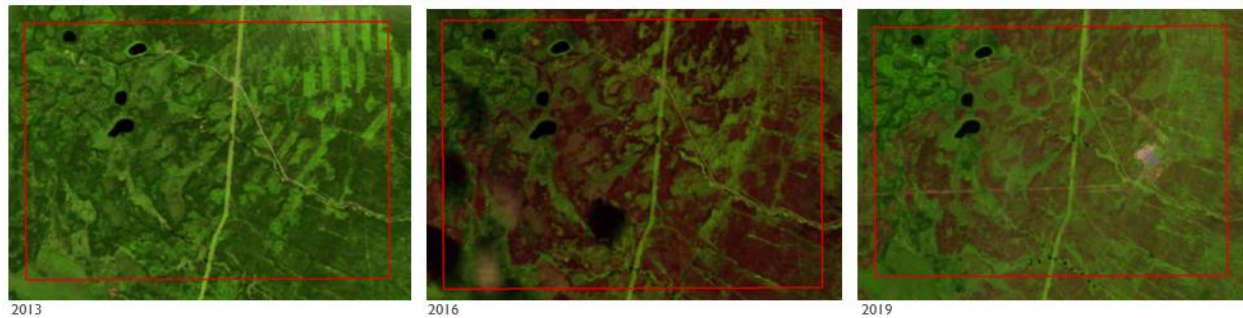


Figure 22: One area of interest to monitor vegetation changes across the 6 years. RGB as 6, 5, 4. Can see significant vegetation loss in 2016 and some vegetation regrowth in 2019.

With this band combination of RGB – 654, we are able to see the changes in vegetation pre and post fire in more detail than the RGB band combination of 543. The use of the SWIR band, we are able to track the direction the fire moved in 2016 and where it was the worst based off of the locations with less vegetation, symbolized by green, even after 3 years of regrowth in the 2019 image (Figures 23 and 24).

Normalized Difference Vegetation Index

NDVI can be used to identify the density of the vegetation in the study as well as the vegetation health. The resulting images in 2013, 2016 and 2019 (Figure 25), display healthier denser vegetation in green, displaying the higher, more positive, NDVI values closer to 1 while the sparse vegetation, soil and bare rock are identified with the yellow in the images as a result of having lower NDVI values, closer to zero or negative numbers. In all images, the red is water or the clouds. The clouds will always present as lower NDVI values. In the 2016 image, some of the values on the bottom right of the image can be excluded from analysis since the cloud presents as lower NDVI values, coloured orange and red. Between the three images, there is more green in the 2013 image indicating a higher quantity of healthy vegetation compared to the other two images where there is more yellow, indicating the presence sparse vegetation and bare soils and rocks exposed post fire. In the 2019

image, there is still significant yellow in the image, however there is more green than in the 2016 image, signaling vegetation regrowth in the area that is returning NDVI values around 0 – 0.2.

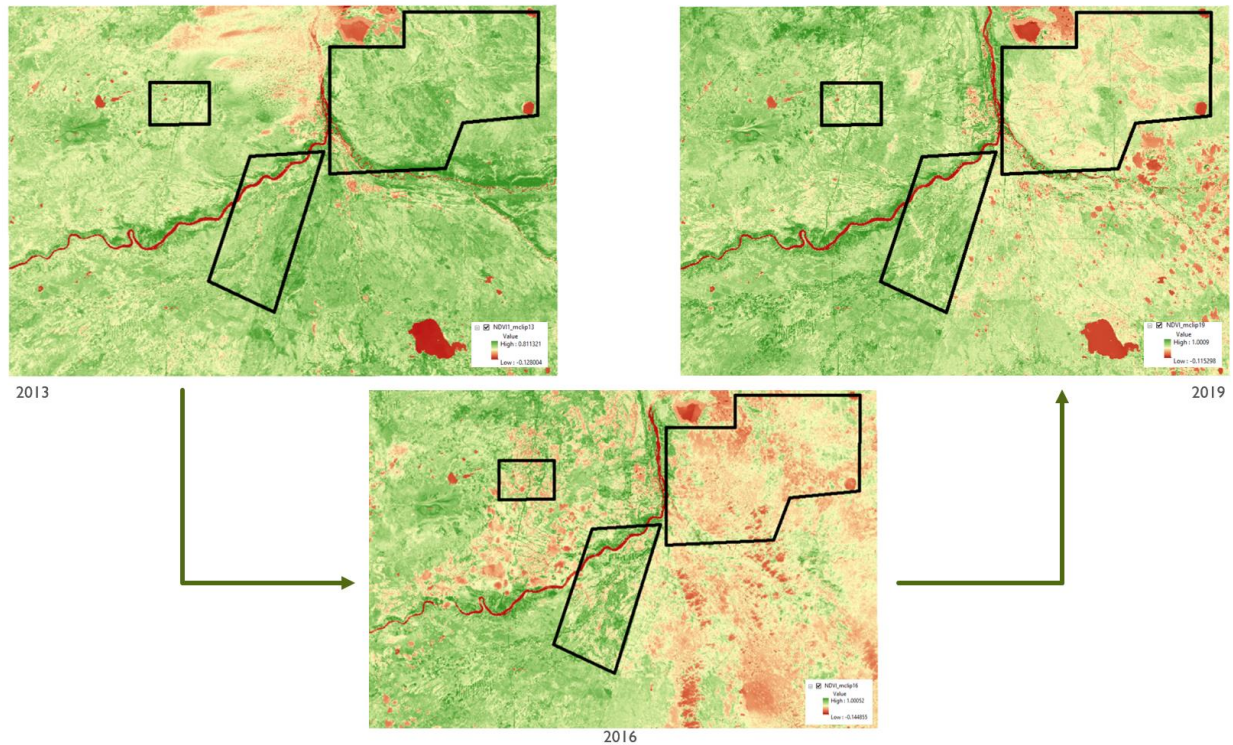


Figure 23: The resulting NDVI images for 2013, 2016 and 2019. The green indicates positive NDVI values, the yellow around 0 and the oranges and reds are negative NDVI values. The black boxes are the areas of interest to track the change across the six years.

Using the classification method in ArcMap to identify intervals to classify the data by, the resulting histogram can be used to examine the distribution of the NDVI values (Figure 26). In 2013 (Figure 26 - 2013), the majority of the values are within 0.2 and 0.3, with more above 0.3. In 2016 (Figure 26 - 2016), the values are more spread out across the x axis, between 0.09 and 0.3. One issue with this classification is that the cloud in the bottom right corner of the image needs to be considered, which can skew these values lower, shifting the histogram to have more values closer to -1. In 2019 (Figure 26 - 2019), there are less values between 0.2 and 0.3, with more below 0.2. Comparing 2013 and 2019, both histograms have the same curve however the 2019 one is shifted towards the lower values, indicating a lower presence of healthy and dense vegetation. In the images, within the black circle, the difference in NDVI indicated by the histogram can be seen. In 2013 there is an abundance of

green (NDVI values of 1 – 0.3 and 0.3 – 0.2) indicating dense, healthy vegetation compared to 2016 and 2019. In 2016, this section within in the circle is not affected by cloud cover, there is more yellow (NDVI values of 0.09 – 0.3 and 0.09 - - 0.05), indicating the presence of sparse vegetation and bare soil and rock as a result of the fire burning through the area. In 2019, there is still yellow (NDVI values of 0.2 – 0.09), however there is more light green in the circle, returning more NDVI values between 0.3 – 0.2. This indicates that there is vegetation regrowth in the area but the vegetation regrowth has not yet reach the levels of the 2013 pre fire vegetation.

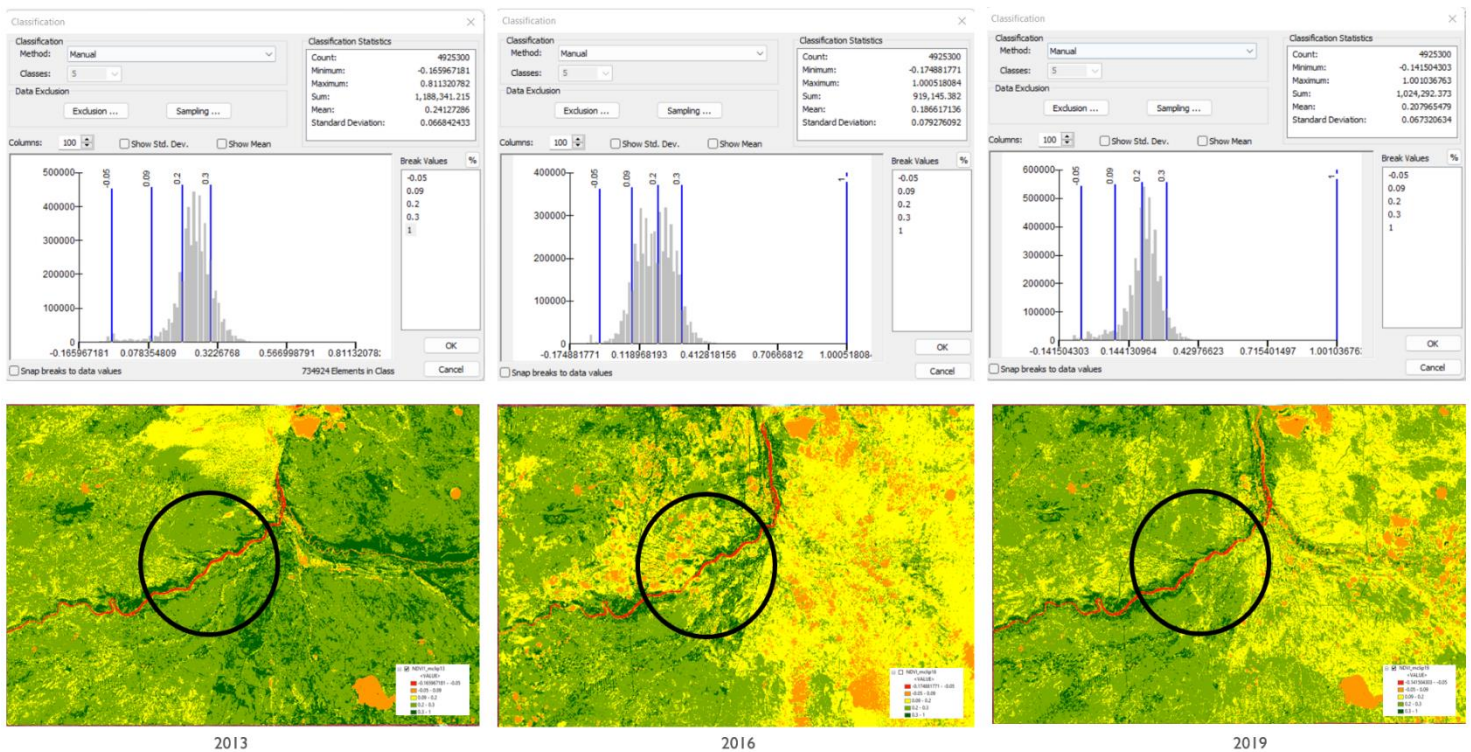


Figure 24: The histograms displaying the distribution of NDVI values within the image. The intervals are highlighted with the blue lines. The resulting images are below with the black circle indicating the are to focus on to identify the change in NDVI across the six years.

Comparing the stretched and classified method of displaying the NDVI values (Figure 27), focusing on the area in the top left of the image where there is no cloud cover in all images, both show the same trend of vegetation loss immediately post fire in 2016 and the start of vegetation regrowth three years post fire in 2019. The stretched classification displays the values in a more continuous manner while the classified classification displays defined classes. Both are ideal for this type of study, however using the classified classification, certain NDVI values can be isolated more easily for studies.

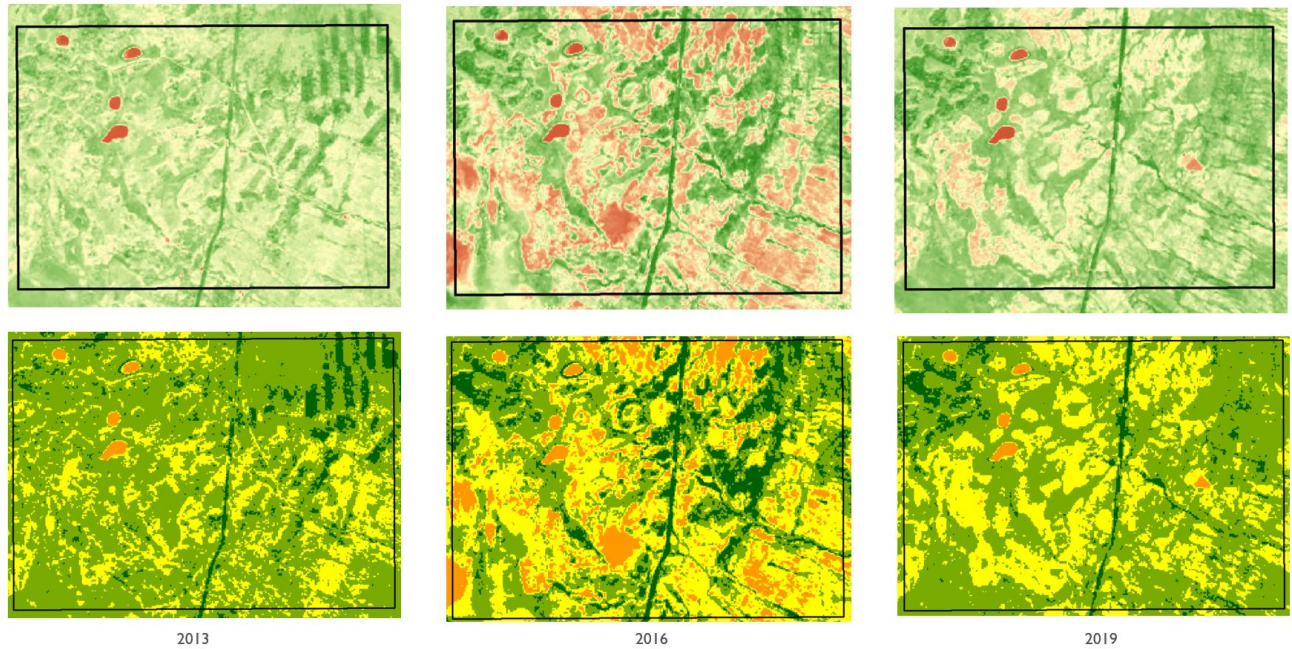


Figure 25: The area of interest in the top left of the larger image. In this section there are significant changes with the NDVI values. Both classification schemes display the same information, the classified version with intervals just has stronger boundaries. Green is higher NDVI values, yellow is values around zero, oranges and red are negative NDVI values.

Normalized Burn Ratio

By applying the NBR equation to the image, we are able to determine the areas that were burned. In Figure 28, the NBR is displayed for 2016 and 2019 in a purple – brown colour scale, with purple indicating positive values up to +1 and brown indicating negative values to -1, the areas with burns. In 2016 we can see significant areas with negative values, coloured brown in the image, identifying the areas burnt the worst from the fire. There is the cloud in the right half of the image, so to focus on the change we will use the left half of the image, north of the river for comparison with the 2019 NBR. In 2019, we can see that there are still strong areas that were burnt, however there are more areas covered in light to dark purple, indicating increasing positive NBR values, identifying areas that are no longer as burnt, indicating vegetation regrowth.

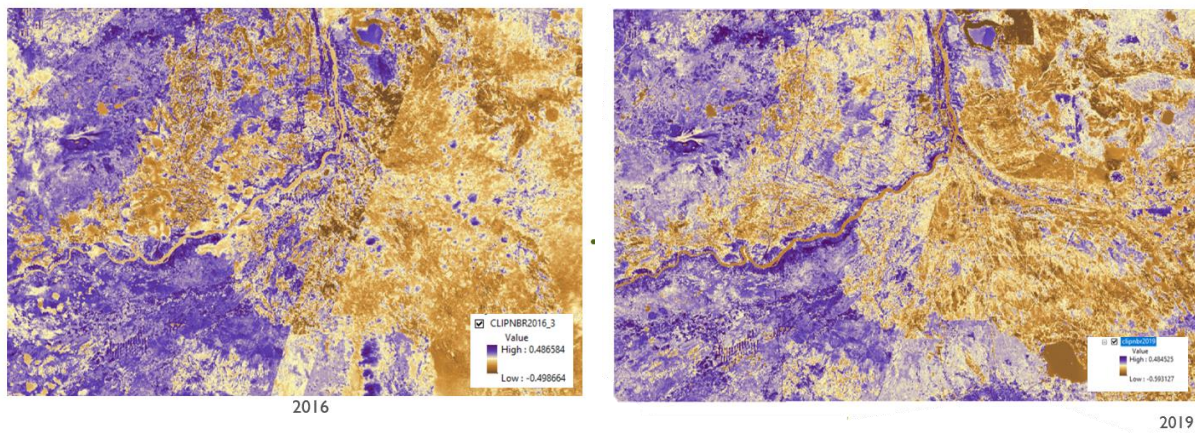


Figure 26: The NBR images for 2016 and 2019. Brown indicates the areas with burn scars. In 2016 we can ignore the bottom right of the image due to the cloud located there in the image.

Difference Normalized Burn Ratio

Using the pre and post fire NBRs, the Difference Normalized Burn Ratio, produces an image that calculates the burn severity. With the values of the DNBR displayed using the purple-brown colour scale, the areas of severe burn are indicated in purple, with brown indicating areas with vegetation. This can be used to determine the locations where the fire was the worst using the DNBR between 2013 pre fire NBR and the 2016 immediately post fire NBR. For the resulting DNBR 2013 – 2016 (Figure 29), due to the cloud in the bottom right of the image, we can exclude that section from our analysis however on the left-hand side of the image, there is a significant section with severe burns, indicated in purple. The location of the severe burned area matches the direction the fire moved North West in later of its life span after May 16. 2016.

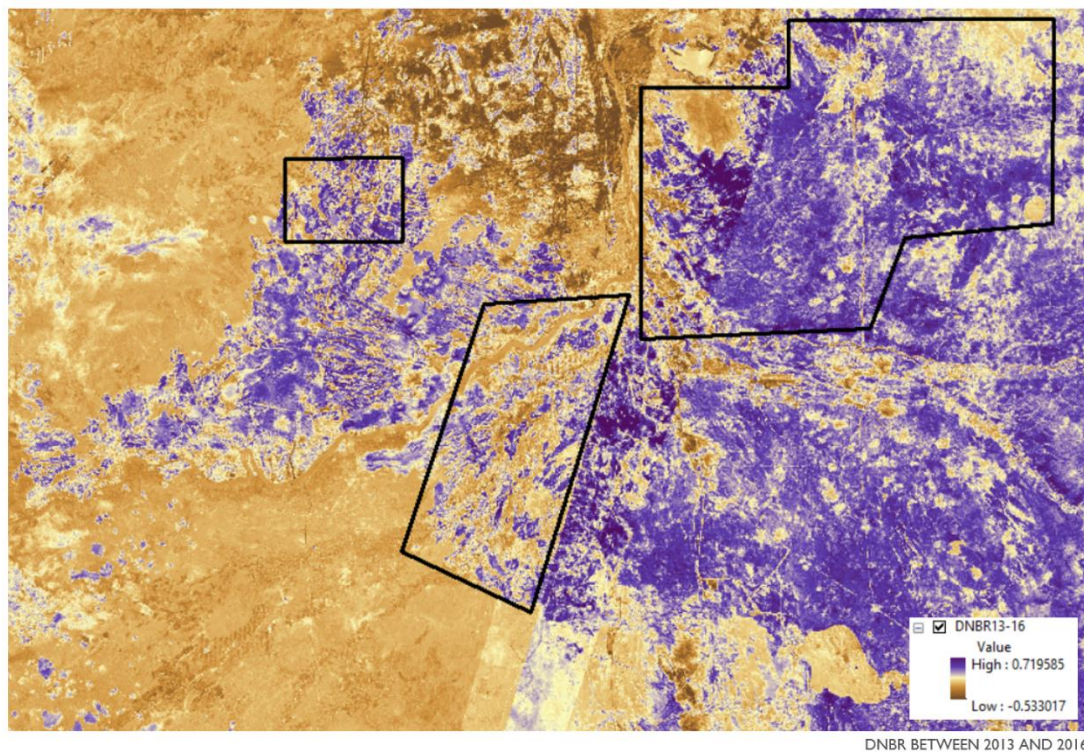


Figure 27: The DNBR between 2013 and 2016. Purple indicates areas with severe burns. The black shapes are areas to use to identify changes throughout the six years. Can exclude the bottom right of the image from analysis due to the cloud cover.

The DNBR between the 2013 pre fire NBR and the 2019 post fire NBR (Figure 30) has significant severe burned areas in the right-hand side of the image, indicated in a dark purple. The most severe burns are in this area which correspond with the start of the Horse River fire and the direction the fire first spread (as seen in figure 29 above). In the top left of the image there are sections indicating severely burnt areas, however there are areas in light yellow, indicating DNBR values around zero, identifying areas with vegetation regrowth.

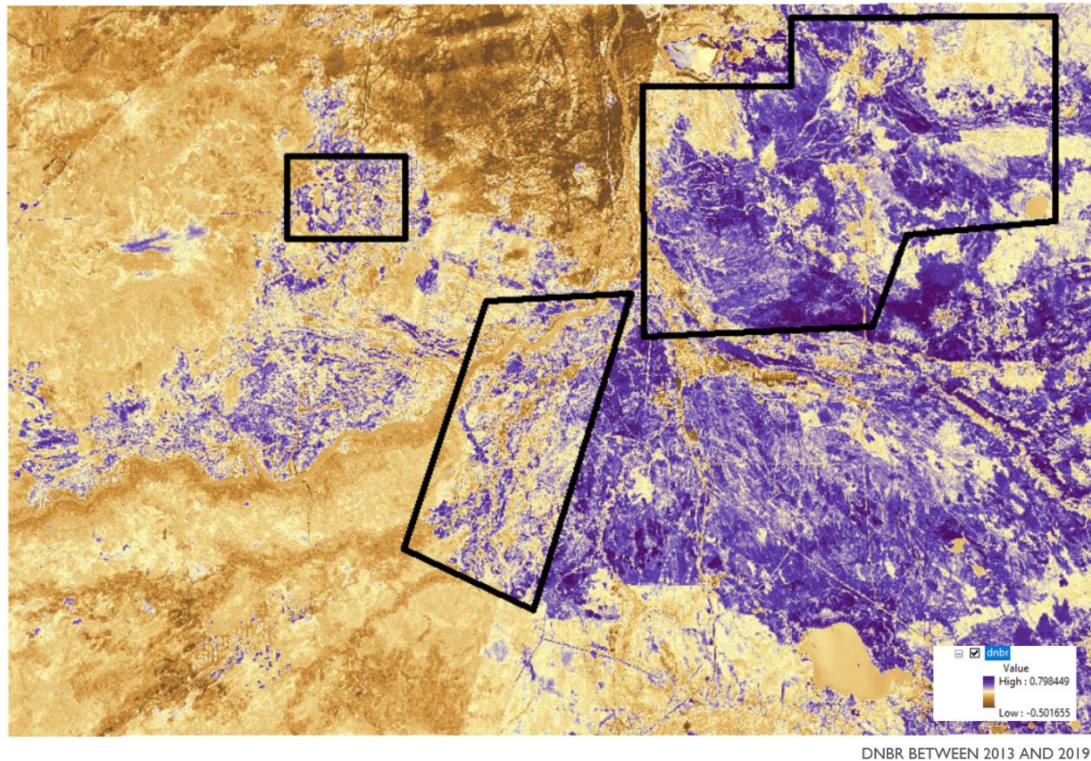


Figure 28: The DNBR between 2013 and 2019. Purple indicates areas with severe burns. The black shapes are areas to use to identify changes throughout the six years.

Comparing the DNBR 2013 – 2016 (Figure 29) and the DNBR 2013 – 2019 (Figure 30), excluding the right hand side of the image, there is significantly more severely burned areas in the DNBR 2013 – 2016 than the DNBR 2013 – 2019. With the DNBR 2013 – 2016, the increase in values close to are zero indicate that there has been vegetation regrowth in the areas three years post fire and the areas of severe burn are no longer as large, decreasing the return of negative DNBR values. The values are moving towards zero and displayed as a light purple. Focusing on the rectangle on the top left of both DNBR images, since this location had no clouds in both images, the change in the areas of severe burn can be

compared in more detail (Figure 31). The change in the severe burned areas decreasing with the DNBR 2013 – 2016 can be linked to the start of vegetation regrowth in the area.

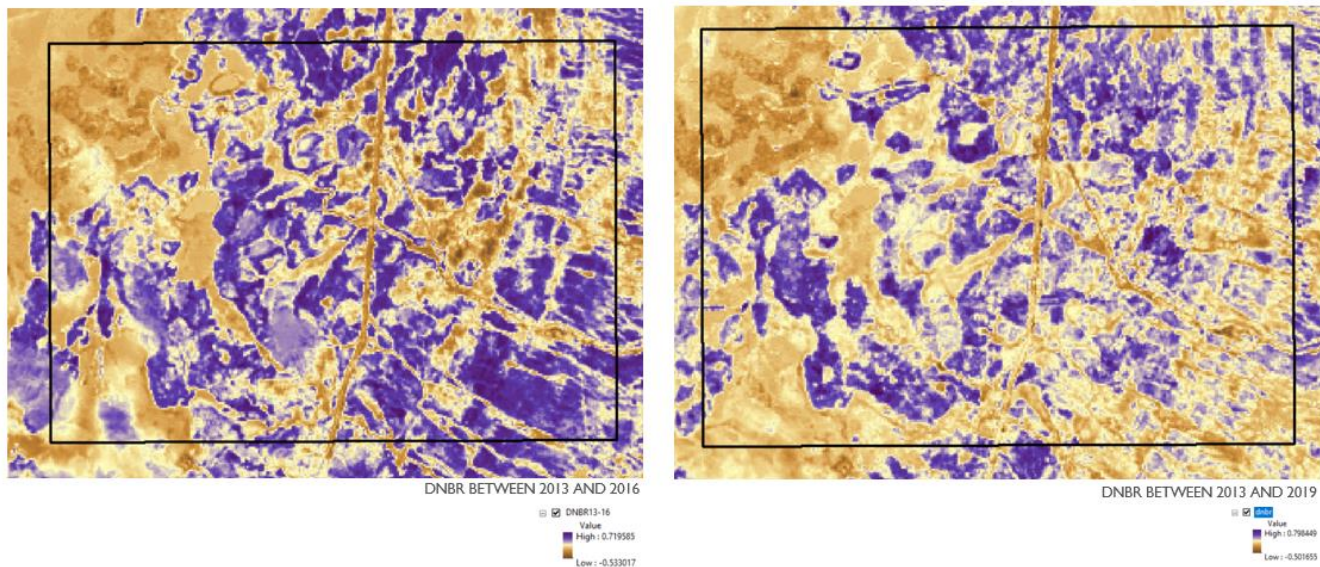


Figure 29: Comparison of the region in the top left of the DNBR images, DNBR 2013 - 2016 on the left, DNBR 2013 - 2019 on the right. Purple indicates areas with severe burns.

Through NDVI and DNBR the changes in vegetation within the study area can be identified from 2013 to 2019. Combining the results, the vegetation is sparser in areas where the burn severity is the worst, with vegetation rebounding the least in those areas. The areas where the NDVI was within the 0.3 to 0.09 range showed to be more affected by the fire as the burn severity calculated through DNBR was the strongest in those sections. This indicates that the sparser vegetation was more susceptible to the fire whereas the denser vegetation, while still affected, was not completely destroyed, still providing leaves for satellites sensors to pick up their reflectance values.

The pattern with vegetation changes and burn severity from the Horse River Fire is similar to the vegetation changes post fire after the Euborea Island in Greece (Figure 32). Even though the overall vegetation type differs, using NDVI and DNBR together can provide an in depth look at the forest fire effects. The areas more severely burnt by the fire in Greece, determined through DNBR, had slower vegetation regrowth than those not as affected, determined through the difference of the pre and post fire NDVIs.

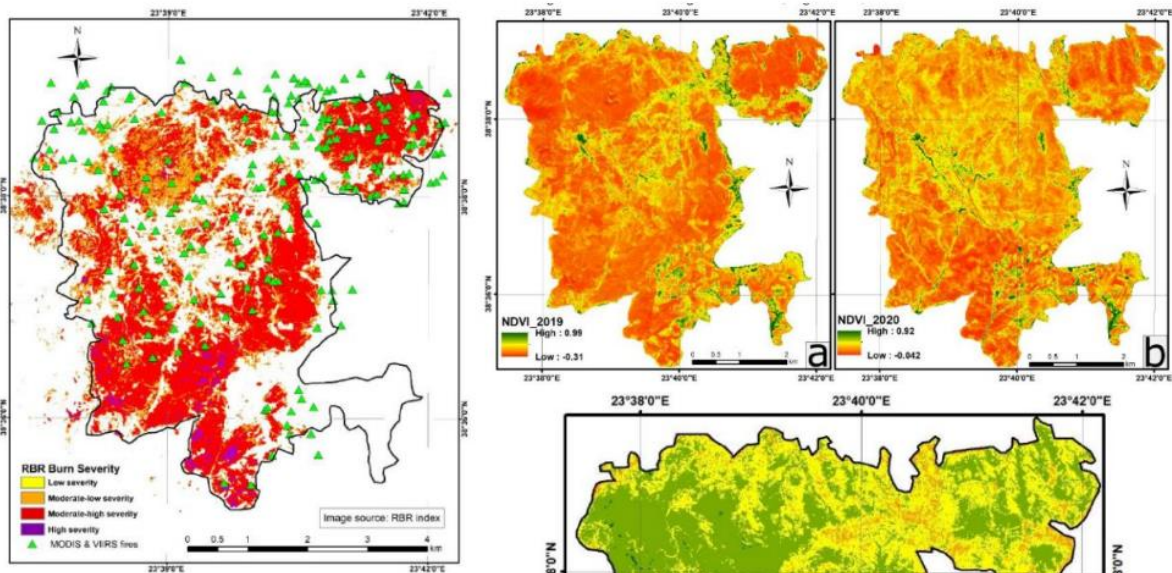


Figure 30: Analysis of the wildfire impacts on vegetation on Euboea Island in Greece [19].

Top Left – map depicting the severity of the burn left over from the wildfire. The burn is worse in the south since that is where the fire started. Top Right – a depicts the vegetation NDVI rating before the wildfire and b is after the fire. Can see that overall, after the fire there is less high-density vegetation. Bottom Right – shows the change in vegetation from before and after the fire. It indicates areas where there is growth and loss. There is moderate vegetation regeneration in the south versus high regrowth levels in the north.

Combining all together we can determine that for this area there is significant amounts of vegetation regeneration after the fire. Can notice that it is slow regrowth in the south where the worst of the burns from the fire are located. Next, can use this model to determine the average vegetation generation per year to creating a timeline for the areas returning to its original vegetation levels.

This data can then be used to look at the areas with vegetation regrowth and determine if natural growth is sufficient or human intervention will be needed to aide in the recovery of vegetation density to pre fire levels.

Future Work

Building off the study on vegetation health and density pre and post Horse River Fire, the next steps would be to take that information and apply it to a land cover analysis of the area. Using the land cover data, the vegetation types in the study area can be identified and determine which tree type or plant was burnt the most by the fire, which remained, which has regrown or if new species have arrived in the area post fire. The change in vegetation types combined with NDVI can help to understand the effects of fire on vegetation regrowth of certain species within the boreal forest and applied to other locations, helping to prepare for future fires in high risk areas of boreal forests.

There was a study completed in a region of the Alaskan Boreal Forest to determine which tree types burn the worst and rebounded the quickest after forest fires. The study concluded that Aspen trees were the least affected and if there was an abundance of deciduous trees, the fire would be less severe and burn more slowly [18]. It would be interesting to see if the results of the study done in the boreal forest around fort McMurray yielded similar results or if being further south affects the outcome.

Next would be to calculate the difference in the pre and post fire NDVIs to obtain a more concrete values for areas with strong or weak vegetation regrowth similar to a study done after the forest fire on Euboea Island in Greece (Figure 32). This would determine the exact locations with strong vegetation regrowth and can be used to identify which types of plants and trees are growing more quickly post fire and how they are changing the post fire ecosystem.

Issues and Errors

With this study there some issues encountered with the data as well as software used. In the 2016 image for immediately post fire, within the study area there is a cloud covering the majority of the bottom right hand corner. This unfortunately affected the NDVI, NBR and DNBR analyses as the cloud cover will be classified within a range that it should not actually be classified in. For example, for NDVI, the cloud is classified within the lower values, which can be misinterpreted as less dense, sparse vegetation or bare rock and soil. To correct this, an image will need to be downloaded from the USGS Earth Explorer with less cloud coverage in that area. Unfortunately, the original image chosen and used contained the least cloud cover for the time frame of the study compared to the other options available.

To process and analyse the data, ENVI was the first chose of software to use. Unfortunately, the version of ENVI available was not able to read the Landsat 8 acquired for the study area so ArcMap was used instead. The software ArcMap worked for the majority of the study but did not offer the option to do a statistical analysis. When visually identifying the change through RGB band combinations and the NDVI analysis the next step would have been to compare the histograms for bands 1 – 7, specifically bands 4, 5, 6 (red, NIR and SWIR) to see the changes in values between the pre and post fire data to statistically determine vegetation changes as well as analyse the changes using the spectral profiles of various pixels against the standard reflectance values (Figure 33). For this section of the study, ENVI would have been useful.

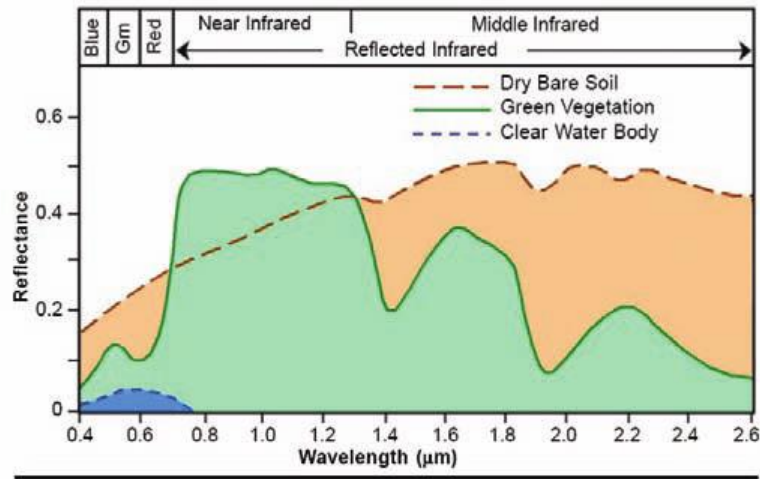


Figure 31: The reflectance responses of water, vegetation and soil for the blue, green, red, NIR and IR wavelengths [20].

Conclusion

With this study, after the Horse River fire there was significant vegetation loss. Following the path of the fire, the vegetation health and density significantly decreased in the areas where the burn was most severe. The vegetation density has increased in the three years post fire from 2016 to 2019, indicating that regrowth has begun. Comparing 2013 and 2019, the post fire vegetation levels have not returned to the pre fire levels, however the areas affected by severe burns has decreased.

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